ZS-F15

Time-Rotation Co-Definition: The Origin of First Rotation as Planck-Unit-Establishing Event

Closure of NC-M12.4 via Mexican-Hat Stage and A-Holonomy Torque; FFPP Sixth Axiom Extension

Kenny Kang
Z-Spin Cosmology Collaboration
April 2026 - ZS-F15 v1.0(Revised) (Foundations Theme) | Paper 15 of 15

Verification: 35/35 PASS | Zero New Free Parameters | Anti-Numerology STRONG PASS

§0. Abstract

ZS-M12 v1.0 §10 NC-M12.4 records a structural assumption that has remained at the foundation of the Z-Spin cyclic cosmology chain: "Q = A assumes $\dot{\theta} = A$ at Z-Telomere onset." Both the comoving U(1) charge $Q = a^3\varepsilon^2\dot{\theta}$ and the geometric impedance A = 35/437 are PROVEN, but the angular velocity $\dot{\theta}(0)$ at the onset of the first rotation has been carried as an axiom rather than a derived quantity. This paper closes NC-M12.4 by recognizing two complementary structures: (i) the Mexican-Hat potential V(Φ) provides the stage on which rotation becomes possible (the S^1 vacuum manifold opens the angular Goldstone direction); (ii) the geometric impedance A acts as a Wilson-loop holonomy $\oint \omega = A$ on the polyhedral defect manifold, providing the torque that biases the first phase drift. Time and rotation are co-defined at the Mexican-Hat Bootstrap Event by a single unit-fixing identity $\dot{\theta}(0) \cdot \tau_P = \delta\varphi_cell = A$.

The Stage-Torque Duality is the principal new structural insight of v1.0(Revised). The Mexican-Hat stage (V(Φ) symmetric under U(1)) by itself opens the rotation direction but cannot bias it; the A-holonomy torque ($\oint \omega = A$, fixed by Gauss-Bonnet on the polyhedral defect manifold per ZS-F3 §6 + ZS-F4 §3-§6 DERIVED) provides the directional bias. The same A-holonomy that produces the per-cycle phase drift $\delta\varphi_cell = A$ is also responsible for the macroscopic Hubble ratio $H_0_local/H_0_CMB = \exp(\oint \omega) = \exp(A)$ (ZS-F3 §6 DERIVED), establishing the polyhedral holonomy as the unifying mechanism across Planck and cosmological scales.

The pre-existence state $|\Phi| = 0$ (the Mexican-Hat ridge of $V(\Phi) = (\lambda/4)M_P^4(|\Phi|^2 - 1)^2$, ZS-F1 §3.3 PROVEN) is a Y-only world: the angular Goldstone θ is undefined when the radial mode ε vanishes (Z-Anchor theorem, ZS-F1 §5.2 PROVEN; ZS-A6 §4.4 DERIVED), so no time evolution is operationally meaningful. The 6 of 11 register slots populated by Y-sector modes (gravitational entropy modes, ZS-Q7 §6.4 PROVEN) carry the entire pre-rotation information content. The first rotation event is the spontaneous symmetry breaking $|\Phi|: 0 \to 1$ that simultaneously selects a phase $\theta_1 \in [0, 2\pi)$ (Goldstone

vacuum manifold, $\pi_1(U(1)) = \mathbb{Z}$ PROVEN), establishes the Planck time τ_P, and acquires the holonomy phase drift $\delta\varphi_{cell} = A$ from the A-holonomy torque.

Five theorems are established. Theorem F-rotor.1 (Y-Only Pre-Existence State, DERIVED) identifies the pre-rotation state as operationally time-free with Y-sector dominance. Theorem F-rotor.2 (Mexican-Hat Bootstrap Event, DERIVED) describes the $|\Phi|$: $0 \to 1$ transition as a single dynamical event simultaneously instantiating radial mode, angular mode, and time evolution. Theorem F-rotor.3 (Time-Rotation Co-Definition, DERIVED — central new contribution) establishes $\theta'(0) \cdot \tau_P = \delta\varphi_{cell} = A$ as a unit-fixing identity rather than a free parameter. Theorem F-rotor.4 (FFPP Sixth Axiom Extension, HYPOTHESIS-strong) extends the FFPP minimal axiom set by an explicit sixth axiom (Time-Rotation Co-Definition). Theorem F-rotor.5 (Stage-Torque Duality, DERIVED — new in v1.0(Revised)) formalizes the separation between Mexican-Hat stage (enables rotation, U(1)-symmetric) and A-holonomy torque (biases first phase drift, fixed by Gauss-Bonnet). Lemma F-rotor.6 (Goldstone Long-Distance Survival, DERIVED — new in v1.0(Revised)) connects the first rotation to galactic-scale halo structure via the angular Goldstone propagation.

The framework aligns with five independent external lines from quantum gravity. Loop quantum gravity (Rovelli 2004, Ashtekar 2021) predicts Planck time as emergent from spin-network dynamics. Hartle-Hawking no-boundary cosmology (1983) proposes chronogenesis as a phase transition. The Page-Wootters formalism (1983, with 2024-2025 reinterpretations Smith-Ahmadi 2019, Mendes et al. 2021, MDPI Universe 10 (2024) 73) establishes that time emerges from quantum correlations between a clock subsystem and a system of interest, in close structural alignment with the Time-Rotation Co-Definition. Geometric Baryogenesis with Chiral-Time Equivalence (Pradisi et al. 2026, arXiv:2604.03973) postulates a diagonal transformation mixing time diffeomorphism with axial U(1) rotation. Pre-geometric gravity (Addazi et al. 2026) generates spacetime metrics through spontaneous symmetry breaking from a pre-geometric theory.

Outlook §11.5 registers a STRUCTURAL INSIGHT candidate: the inflationary tensor-to-scalar ratio $r = 0.0089$ (ZS-U1 §4.2 DERIVED + TESTABLE) may be the macroscopic tensor imprint of the first rotation event, with the same A-holonomy structure that produces $\delta\varphi_{cell} = A$ at Planck scale producing the tensor power amplitude at horizon-exit scale via the slow-roll formula $r = 16\varepsilon_V$ with ε_V depending on A through the Jordan kinetic term dominance $1/(6A)$. LiteBIRD (~2032+) will test this connection at ~6σ via the discriminator $r_{ZS}/r_{Starobinsky} = 2.67$. Six new falsification gates F-F15.1 through F-F15.6 are pre-registered. Verification: 35/35 PASS across 9 test categories. Zero new free parameters; the contribution is structural reinterpretation, not new numerical content.

Keywords: first rotation origin, time-rotation co-definition, Mexican-Hat stage, A-holonomy torque, Stage-Torque Duality, Y-only pre-existence, Planck time emergence, NC-M12.4 closure, FFPP sixth axiom, Page-Wootters formalism, axion kinetic misalignment, chiral-time equivalence, Wilson loop, Gauss-Bonnet polyhedral defect manifold, Goldstone phase selection, primordial tensor imprint, zero new parameters.

Epistemic Status Legend

STATUS	DEFINITION
LOCKED	Core constant derived and fixed in upstream paper; no downstream paper may modify.
PROVEN	Mathematical theorem with complete proof; verified to machine or 50-digit precision.
DERIVED	Quantitative consequence of PROVEN items combined with Z-Spin axioms; zero free parameters.
DERIVED-CONDITIONAL	Derived from Z-Spin axioms conditional on a stated upstream assumption.
VERIFIED	Numerical confirmation against observational data or independent computation.
TESTABLE	Quantitative prediction with pre-registered falsification condition.
HYPOTHESIS-strong	Multiple independent structural lines of evidence; full chain incomplete.
STRUCTURAL INSIGHT	Reinterpretation of existing PROVEN content under a new conceptual frame.
NON-CLAIM	Explicit declaration of what this paper does NOT establish.
OPEN	Identified gap or sub-computation pending future work.

§1. Introduction

§1.1 The NC-M12.4 Open Item and the Stage-Torque Question

ZS-M12 v1.0 §7.1 establishes the U(1) Noether comoving charge $Q = a^3\varepsilon^2\dot{\theta}$ at the Z-Telomere onset, with $a = 1$, $\varepsilon = 1$ fixed by the conditions of the bounce, and $\dot{\theta}$ taken as $A = 35/437$ by reference to ZS-U5 v1.0 Lemma 8.1 ($\delta\varphi = A$ radians per Regge cycle, DERIVED-under-P6, with P6 itself closed via Theorem P6* in ZS-U8 §3.0a). The cyclic cosmology chain therefore reads:

$$A = 35/437 \rightarrow \delta\varphi_cell = A \rightarrow Q_initial = A \rightarrow \varepsilon_min(a=1) \approx 30.7 \rightarrow \text{ZS-U1 inflation IC}$$

However, NC-M12.4 (ZS-M12 v1.0 §10) records an explicit caveat: "$Q = A$ assumes $\dot{\theta} = A$ at Z-Telomere onset. Modification of $\delta\varphi$ changes ε_min." The non-claim concedes that $\dot{\theta}(0) = A$ is taken as input rather than derived from the Z-Spin action.

Closer examination reveals that the question "why $\dot{\theta}(0) = A$?" splits into two logically distinct sub-questions. First, what mechanism opens the rotation direction in field configuration space — i.e., what makes angular accumulation possible at all? Second, what biases the rotation toward a specific phase drift rate — i.e., what fixes the value $\delta\varphi = A$ rather than zero or some other value? We refer to the first as the stage question and the second as the torque question. The Mexican-Hat potential $V(\Phi)$ provides the stage: its U(1) symmetry creates a flat angular direction in the vacuum manifold (Goldstone direction), opening rotation as a possibility but not specifying a value. The geometric impedance A acts as a Wilson-loop holonomy on the polyhedral defect manifold, providing the torque: by Gauss-Bonnet on the X-sector (truncated octahedron, O_h) crossed with the Y-sector (truncated icosahedron, I_h), $\oint \omega = A$ is fixed (ZS-F3 §6 DERIVED + ZS-F4 §3-§6 PROVEN). This Stage-Torque Duality is the principal new structural insight of v1.0(Revised); it is formalized as Theorem F-rotor.5 in §6.5.

§1.2 The Y-Only World Insight

The structural origin of the NC-M12.4 caveat becomes clear once the pre-rotation state is examined. Before the first cycle, the field configuration is $|\Phi| = 0$ — the Mexican-Hat ridge of the Z-Spin potential $V(\Phi) = (\lambda/4)M_P^4(|\Phi|^2 - 1)^2$ (ZS-F1 §3.3 PROVEN; ZS-A6 §1.3 Definition Lock 2 corrects the misnomer "false vacuum"). At $|\Phi| = 0$, the polar decomposition $\Phi = \varepsilon e^{i\theta}$ fails: the angular coordinate θ is mathematically undefined when the radial coordinate ε vanishes. The Z-Anchor theorem (ZS-F1 §5.2 PROVEN, restated in ZS-A6 §4.4.3 Component 1) establishes this as a topological necessity: if the field has any winding number $n \neq 0$ around a point x_0, then $|\Phi(x_0)| = 0$ by single-valuedness. Consequently, the pre-rotation state has no angular dynamics, no operational time, and no specified stage on which rotation can occur.

The 11-slot register $Q = Z + X + Y = 2 + 3 + 6$ (ZS-F5 PROVEN) reduces operationally to its Y-sector content (gravitational modes, dim = 6, ZS-Q7 §6.4 PROVEN identification): $p_{eq} = (3, 2, 6)/11$ with the Y component carrying $6/11 \approx 54.5\%$ of the entropy budget. The Z-sector (dim = 2, mediator) and X-sector (dim = 3, matter) are present in the action but operationally inert until the rotation begins. This is the structural realization of what we will call the "Y-only pre-existence": time-free, rotation-free, Y-dominated. The first rotation event is the moment at which both the stage (Mexican-Hat S^1 vacuum manifold) and the torque (A-holonomy $\oint \omega = A$) become operationally activated, with their joint action establishing the Planck time unit.

§1.3 What This Paper Establishes

This paper establishes: (i) a structural reinterpretation of NC-M12.4 from "assumed parameter" to "unit-fixing identity, properly characterized" (Theorem F-rotor.3); (ii) a derivation of the Y-only pre-existence state as the operational starting point of the cyclic cosmology, using only PROVEN/DERIVED corpus inputs (Theorem F-rotor.1); (iii) a structural identification of the Mexican-Hat rolldown $|\Phi|: 0 \to 1$ with the simultaneous instantiation of radial mode, angular mode, and time evolution (Theorem F-rotor.2); (iv) a proposed extension of the ZS-F0 v1.0(Revised) §13 Fundamental Fixed Point Principle by an implicit sixth axiom (Time-Rotation Co-Definition) that all Z-Spin numerical derivations have been silently using (Theorem F-rotor.4); (v) the Stage-Torque Duality (Theorem F-rotor.5) — the central conceptual decomposition of the first-rotation problem into two physically distinct mechanisms; (vi) the Goldstone Long-Distance Survival lemma (Lemma F-rotor.6) connecting the first rotation to galactic-scale halo structure via ZS-A1; (vii) cross-references to five independent external quantum gravity frameworks that converge on time-rotation co-definition; (viii) an outlook section §11.5 registering $r = 0.0089$ as a candidate macroscopic tensor imprint of the first rotation.

This paper does NOT: (i) derive $\theta(0) = A$ from a smaller axiom set than the existing five-axiom set $\mathscr{A}$ of FFPP (the principal new content is the explicit identification of the implicit sixth axiom and the Stage-Torque Duality, not a reduction in axiom count); (ii) introduce a new numerical prediction (no new observable quantity is computed); (iii) resolve the Mexican-Hat phase selection mechanism (the choice of $\theta_1 \in [0, 2\pi)$ at SSB remains spontaneous; partial chiral biasing from Y-sector I_h structure is registered at HYPOTHESIS-strong); (iv) derive the Planck time in physical units (the unit-fixing identity is internal

to Z-Spin natural units); (v) introduce any new free parameter beyond A = 35/437. The advance is structural (reinterpretation + decomposition), not numerical.

ZS-F15 sits in the Foundations theme as Paper 15 (the closing paper of the F-theme). It draws principally on ZS-F0 v1.0(Revised) (FFPP, ontological bootstrap), ZS-F1 (Z-Spin action, Mexican-Hat potential), ZS-F3 (holonomy $\oint \omega = A$, exp(A) Hubble ratio derivation), ZS-F4 (Wilson loop on polyhedral defect manifold), ZS-M12 (Auto-Surgery, U(1) Noether charge, NC-M12.4 itself), ZS-U5 (Regge-Holonomy, $\delta\varphi_cell = A$), ZS-U11 (four-channel Q-protection, conjugate decomposition), ZS-A1 (Goldstone θ-mode long-distance halo dynamics), ZS-Q7 (Y-sector gravitational entropy dominance), and ZS-U1 (inflationary observables r = 0.0089). All inputs are LOCKED, PROVEN, or DERIVED in those prior papers.

§1.4 Changes from v1.0 to v1.0(Revised)

v1.0(Revised) adds five substantive enhancements to the v1.0 manuscript while preserving all v1.0 content verbatim per the Z-Spin no-deletion rule. (a) Theorem F-rotor.5 (Stage-Torque Duality, §6.5): explicit decomposition of the first-rotation problem into two physically distinct mechanisms — Mexican-Hat stage and A-holonomy torque. (b) Lemma F-rotor.6 (Goldstone Long-Distance Survival, §3.5): explicit integration with ZS-A1 §2.1 establishing that the angular Goldstone mode survives to galactic scales because of $m_\theta = 0$ exactly. (c) §5.8 explicit holonomy formulation: the identity $\theta^{\cdot} \cdot \tau_P = \delta\varphi = A$ is now expressed in Wilson-loop form $\oint \omega = A$ with explicit Gauss-Bonnet derivation, connecting to the macroscopic Hubble ratio $H_0_local/H_0_CMB = \exp(A)$. (d) §11.5 (Outlook): r = 0.0089 registered as STRUCTURAL INSIGHT candidate for macroscopic tensor imprint of first rotation. (e) Page-Wootters formalism added as fifth external cross-reference (Appendix C.5). The verification suite is extended from 28 to 35 tests; the anti-numerology Monte Carlo from 250,000 to 350,000 trials. All v1.0 numerical content and theorem statements are preserved unchanged.

§2. Locked Inputs

All quantities used in this paper are LOCKED, PROVEN, or DERIVED in prior corpus papers. Zero new parameters are introduced. The Planck time τ_P appears as a unit of time in Planck-natural units; its identification with the first-cycle Regge interval is the structural content of §5 and is the principal new statement of the paper.

Table 2.1. Locked inputs to ZS-F15 v1.0(Revised).

Quantity	Value / Description	Source	Status
A = 35/437	0.080092	ZS-F2 v1.0	**LOCKED**
Q = 11; (Z, X, Y)	(2, 3, 6)	ZS-F5 v1.0	**PROVEN**
V(Φ) Mexican-Hat	$(\lambda/4)M_P^4(\|\Phi\|^2-1)^2$	ZS-F1 v1.0 §3.3	**PROVEN**
Vacuum manifold	$\{\|\Phi\| = 1\} \cong S^1$	ZS-F1 §5.1	**PROVEN**
$\pi_1(U(1)) = \mathbb{Z}$ winding	$n \in \mathbb{Z}$	ZS-A6 §4.4.2	**PROVEN**

Z-Anchor: $\|\Phi(x_0)\|=0$ if $n\neq 0$	Topological necessity	ZS-F1 §5.2	**PROVEN**
m_ρ (radial mode)	$2A\cdot M_P$	ZS-U5 v1.0 §8.4	**DERIVED-COND**
m_θ (Goldstone)	0 exactly	ZS-S3, Goldstone thm	**PROVEN**
$\Box\theta = 0$ (Goldstone EOM)	Massless wave eq.	ZS-A1 §2.1	**DERIVED**
$Q = a^3\varepsilon^2\dot{\theta}$ (Noether)	U(1) comoving charge	ZS-M12 §7.1	**PROVEN**
$\delta\varphi_cell = A$ per Regge cycle	Phase per primitive cell	ZS-U5 Lemma 8.1	**DERIVED**
P6 closure: $\kappa=4$ unique	$I_cell = 4/4 = 1$	ZS-U8 §3.0a Theorem P6*	**PROVEN**
Y-sector gravitational modes	$p_eq\ Y/Q = 6/11$	ZS-Q7 v1.0 §6.4	**PROVEN**
FFPP 5-axiom set $\mathscr{A}$	{Exist, Self-ref, Alg, NonTriv, Unit}	ZS-F0 §13.1	**DERIVED**
$\varepsilon_min = (Q^2/\lambda)^{(1/6)}$	≈ 30.73	ZS-M12 §A.3	**PROVEN**
$\oint \omega = A$ on polyhedral manifold	Wilson loop of scalar connection	ZS-F3 §6, ZS-F4 §3-6	**DERIVED**
$H_0_loc/H_0_CMB = \exp(\oint \omega) = \exp(A)$	Macroscopic holonomy	ZS-F3 §6, ZS-F4 §6	**DERIVED**
$r = 0.0089$, $n_s = 0.9674$	Inflationary tensor/scalar	ZS-U1 v1.0 §4.2	**DERIVED+TESTABLE**
I_h chirality index Δ	$\neq 0$ for irreps 1, 3, 3', 5	ZS-M9 v1.0 §3.5	**PROVEN**
NC-M12.4 (existing)	$\dot{\theta}=A$ assumed at onset	ZS-M12 §10	**NON-CLAIM**

§2.1 Dependencies

Inputs TO this paper: ZS-F0 v1.0(Revised) (FFPP, B0 axiom, 5-axiom set $\mathscr{A}$, three-layer fixed points), ZS-F1 v1.0 (Mexican-Hat potential, U(1) symmetry, Z-Anchor theorem), ZS-F2 v1.0 ($A = 35/437$ LOCKED), ZS-F3 v1.0 ($\oint \omega = A$ holonomy derivation, exp(A) Hubble ratio), ZS-F4 v1.0 (Wilson loop on polyhedral defect manifold, six-step holonomy chain), ZS-F5 v1.0 ($Q = 11$, sectors), ZS-M12 v1.0 (Auto-Surgery, U(1) Noether $Q = a^3\varepsilon^2\dot{\theta}$, NC-M12.4), ZS-U1 v1.0 (inflationary observables $r = 0.0089$, $n_s = 0.9674$), ZS-U5 v1.0 (Regge-Holonomy, $\delta\varphi_cell = A$), ZS-U8 v1.0 (Theorem P6* closure, ε rolldown ODE), ZS-U11 v1.0 (four-channel Q-protection, conjugate decomposition NC-U11.1), ZS-A1 v1.0 (Goldstone θ-mode, isothermal halo profile $\rho_\theta \propto 1/r^2$), ZS-A6 v1.0 (Z-Anchor, $\pi_1(U(1)) = \mathbb{Z}$ winding), ZS-S3 v1.0 (Goldstone theorem application, $m_\theta = 0$ exactly), ZS-Q7 v1.0 (Y-sector gravitational entropy dominance, $p_eq = (3,2,6)/11$), ZS-M9 v1.0 (I_h chirality index, McKay correspondence).

Outputs FROM this paper: NC-M12.4 status reinterpretation; six theorems (F-rotor.1 through F-rotor.6); seven non-claims NC-F15.1 through NC-F15.7; six new falsification gates F-F15.1 through F-F15.6; verification suite extension Categories [U]–[Z+] (35 tests); cross-paper consistency annotations to ZS-M12 §10, ZS-F0 §13, ZS-U11 §10, ZS-U5 Lemma 8.1, ZS-A1 §2.1; Stage-Torque Duality (Theorem F-rotor.5) as the central conceptual decomposition; $r = 0.0089$ macroscopic tensor imprint candidate (§11.5).

§3. Theorem F-rotor.1 — Y-Only Pre-Existence State

§3.1 Statement

Theorem F-rotor.1 (Y-Only Pre-Existence State, DERIVED). The pre-rotation field configuration $|\Phi| = 0$ of the Z-Spin action S_F1 is operationally a Y-only state in the following precise sense: (i) the polar decomposition $\Phi = \varepsilon e^{i\theta}$ fails (θ undefined when $\varepsilon = 0$); (ii) the Goldstone equation $\Box\theta = 0$ has no dynamical content (Z-Anchor enforces $|\Phi| = 0$ if winding $n \neq 0$, and θ is multivalued at any vanishing point); (iii) no time-evolution generator is operationally meaningful in this state, so τ_P is not yet established as a unit; (iv) the 11-slot register $Q = Z + X + Y$ populated by the equilibrium distribution $p_{eq} = (3, 2, 6)/11$ carries $6/11 \approx 54.5\%$ of its content in the Y-sector, identified in ZS-Q7 §6.4 (PROVEN) as the gravitational-entropy modes, yielding the operational designation "Y-only".

Status. DERIVED. Each clause (i)–(iv) follows from PROVEN/DERIVED corpus inputs without new assumptions.

§3.2 Proof of Clauses (i)–(iv)

Clause (i): Polar decomposition fails at $|\Phi| = 0$. Standard complex analysis: for $\Phi \in \mathbb{C}$, the polar form $\Phi = \varepsilon e^{i\theta}$ is well-defined except at $\Phi = 0$, where $\varepsilon = 0$ forces θ to be arbitrary (Conway 1978 §I.2). At $|\Phi| = 0$, the angular coordinate θ is not a well-defined function of the complex point.

Clause (ii): Goldstone equation has no dynamical content. ZS-A1 §2.1 (DERIVED) and ZS-S3 (PROVEN, Goldstone theorem applied to spontaneously broken U(1) of ZS-F1 §3.2) establish $\Box\theta = 0$ for the angular Goldstone mode in the broken phase $|\Phi| = 1$. The equation is a wave equation on the configuration where $\varepsilon \neq 0$ throughout. At $|\Phi| = 0$, by clause (i), θ is undefined, so $\Box\theta$ has no operational meaning. The Z-Anchor theorem (ZS-F1 §5.2 PROVEN; ZS-A6 §4.4 Component 1 DERIVED) sharpens this: any nonzero winding $n \neq 0$ forces $|\Phi(x_0)| = 0$ by the contradiction between single-valuedness of Φ and the multivaluedness of θ at any vanishing point.

Clause (iii): No time-evolution generator. The U(1) comoving charge $Q = a^3\varepsilon^2\dot{\theta}$ (ZS-M12 §7.1 PROVEN) reduces to $Q = 0$ when $\varepsilon = 0$, regardless of $\dot{\theta}$. The standard time-evolution generator on the angular sector vanishes. The radial sector evolves via Klein-Gordon on $V(\varepsilon)$, but at $\varepsilon = 0$ we have $V''(0) = -\lambda M_P^4 < 0$ (ZS-A6 §1.3 Definition Lock 2 PROVEN: $|\Phi| = 0$ is unstable local maximum), indicating instability rather than time evolution.

Clause (iv): Y-sector populates 6/11 of the entropy budget. ZS-Q7 §5.8 (DERIVED) establishes $p_{eq} = (3, 2, 6)/11$. ZS-Q7 §6.4 (PROVEN identification): "The Y-sector (gravitational modes, dim = 6) dominates the entropy budget by factor 2 over the X-sector. This is the Z-Spin geometric realization of Penrose's physical intuition." In the pre-rotation state, X- and Z-sector modes are present in the action but cannot exchange information ($L_{XY} \equiv 0$ PROVEN forces transfer through Z, but Z has no dynamics until rotation begins). The Y-sector dofs are the only operationally accessible content.

§3.3 External Cross-References

Three independent external frameworks support the Y-only pre-existence interpretation. (a) Loop quantum gravity (Rovelli 2004, Ashtekar 2021) describes the pre-Big-Bang state as spin-network configurations with no preferred time direction. (b) Hartle-Hawking no-boundary cosmology (1983) proposes that the universe has no past boundary; the pre-chronogenesis state is a Euclidean four-manifold with no time. (c) Pre-geometric gravity (Addazi et al. 2026) constructs gauge theory without pre-defined spacetime, where metric emerges from spontaneous symmetry breaking; the pre-symmetry-broken state matches structurally with the Z-Spin $|\Phi| = 0$ ridge.

§3.4 Verification (Category [U])

Five tests verify Theorem F-rotor.1. U.1: Polar decomposition at $|\Phi| = 0$ (machine precision verification that θ at $\varepsilon = 0$ is arbitrary). U.2: Goldstone equation undefinedness at $|\Phi| = 0$ (inherits ZS-F1 §5.2 PROVEN). U.3: $V''(0) = -\lambda M_P^4 < 0$ (ZS-A6 §1.3 PROVEN). U.4: Equilibrium distribution $p_{eq} = (3, 2, 6)/11$ verified (inherits ZS-Q7 §5.8 PROVEN). U.5: Y-sector dominance $Y/Q = 6/11 \approx 54.5\%$ (inherits ZS-Q7 §6.4 PROVEN identification).

§3.5 Lemma F-rotor.6 — Goldstone Long-Distance Survival (NEW in v1.0(Revised))

Lemma F-rotor.6 (Goldstone Long-Distance Survival, DERIVED). After the Mexican-Hat Bootstrap Event (Theorem F-rotor.2), the angular Goldstone mode $\theta(x)$ survives to all macroscopic length scales because $m_\theta = 0$ exactly (Goldstone theorem, ZS-S3 PROVEN). The energy density carried by spatial gradients of θ is $\rho_\theta = \frac{1}{2} M_P^2 (\partial\theta)^2 \propto 1/r^2$, which is precisely the isothermal halo profile (ZS-A1 §2.1 DERIVED). The 56-orders-of-magnitude paradox between the heavy radial mode ($m_\rho = 2A \cdot M_P$, Compton wavelength $\sim 10^{-34}$ m) and galactic-scale (~10 kpc) flat rotation curves is resolved by recognizing that the relevant degree of freedom at galactic scales is the exactly-massless angular Goldstone, not the heavy radial excitation. The first rotation $\dot\theta(0) = A$ at Z-Telomere onset propagates outward without dispersion through the massless Goldstone wave equation; cosmological structure on galactic and larger scales is the macroscopic legacy of the first rotation.

Status. DERIVED. Inherits ZS-S3 PROVEN ($m_\theta = 0$ by Goldstone theorem) + ZS-A1 §2.1 DERIVED ($\rho_\theta \propto 1/r^2$ isothermal halo) + ZS-A1 §3.4 DERIVED (ε-Mass Paradox resolution by Goldstone identification). The v1.0(Revised) addition is the explicit identification of the first rotation as the source of all subsequent macroscopic angular structure.

Physical interpretation: the Mexican-Hat Bootstrap Event creates a Goldstone wave at Planck scale with phase rate $\dot\theta(0) = A$ (per Theorem F-rotor.3); this wave propagates outward at the speed of light because $m_\theta = 0$; spatial gradients $(\partial\theta)^2 \propto 1/r^2$ are the isothermal-halo signature observed in galactic rotation curves (flat at large r). The first rotation is therefore not just a Planck-scale event but the originating source of the entire angular Goldstone field configuration, with macroscopic observational consequences. This connects the first-rotation problem to ZS-A1 (Galactic Dynamics) directly: the dark matter halo is the angular Goldstone mode of the first rotation, propagated through the Mexican-Hat U(1) symmetry to all length scales.

§4. Theorem F-rotor.2 — Mexican-Hat Bootstrap Event

§4.1 Statement

Theorem F-rotor.2 (Mexican-Hat Bootstrap Event, DERIVED). The transition $|\Phi|: 0 \to 1$ from the Mexican-Hat ridge $V(0) = (\lambda/4)M_P^4$ to the vacuum manifold $\{|\Phi| = 1\} \cong S^1$ is a single dynamical event that simultaneously instantiates four structures: (a) a definite radial coordinate $\rho = |\Phi| - 1$ oscillating around the new vacuum; (b) a definite angular coordinate $\theta \in [0, 2\pi)$ selected from the previously continuously-degenerate vacuum manifold (spontaneous U(1) symmetry breaking); (c) an operational time evolution generated by Klein-Gordon dynamics on $V_E(\varepsilon)$; (d) the unit identification of one Regge-cycle interval with one Planck time τ_P, by which subsequent angular accumulation is measured.

§4.2 Standard Spontaneous Symmetry Breaking Content

The Mexican-Hat potential $V(\Phi) = (\lambda/4)M_P^4(|\Phi|^2 - 1)^2$ of the Z-Spin action (ZS-F1 §3.3 PROVEN) exhibits the textbook structure of spontaneous U(1) symmetry breaking. In the Einstein frame (ZS-U1 §2 DERIVED), the canonical potential $V_E(\varepsilon) = (\lambda/4)(\varepsilon^2 - 1)^2/(1 + A\varepsilon^2)^2$ has three regimes: (i) Hilltop $\varepsilon \approx 0$: $V_E(0) = \lambda/4$, local maximum; (ii) Vacuum manifold $\varepsilon = \pm 1$: $V_E(\pm 1) = 0$ exactly (ZS-U8 §2.2 PROVEN, Z_2-degenerate); (iii) Plateau $\varepsilon \gg 1$: $V_E \to \lambda/(4A^2)$ (slow-roll inflation scale).

The Bootstrap Event is the dynamical roll from regime (i) to regime (ii). ZS-U8 §6.3 (DERIVED + VERIFIED) provides the numerical evidence: with critical kinetic factor $f_{crit} \approx 1.0002$, the rolldown completes in approximately 34 M_P^{-1} ($\sim 10^{-43}$ s), with instability rate $\mu = 0.1177\ M_P$. The radial mode acquires mass $m_\rho = 2A \cdot M_P$ (ZS-U5 §8.4 DERIVED-CONDITIONAL) and oscillates around the new vacuum. The angular mode θ becomes a Goldstone boson with $m_\theta = 0$ exactly (ZS-S3 PROVEN). External cross-reference: matches structurally with axion-like models where the angular mode acquires nonzero velocity through misalignment with explicit symmetry-breaking direction (Co et al. 2024, arXiv:2310.17710). The Z-Spin difference is that the explicit symmetry-breaking source is the polyhedral asymmetry $A = \delta_X \cdot \delta_Y$ rather than a phenomenological term.

§4.3 Phase Selection at the Bootstrap Event

The angular phase $\theta_1 \in [0, 2\pi)$ is selected at the moment of $|\Phi|: 0 \to 1$ transition. In the absence of any explicit U(1)-breaking term in the action, the choice of θ_1 is spontaneous: any direction in the vacuum manifold S^1 is energetically equivalent. This is standard Goldstone physics; the Z-Spin framework does not modify it.

However, the Z-Spin Mexican-Hat is not in flat spacetime. The non-minimal coupling $(1 + A|\Phi|^2)R$ of the Z-Spin action (ZS-F1 §3 PROVEN) introduces a curvature backreaction during the rolldown, which biases the phase selection through the parity-asymmetric Y-sector chiral structure (ZS-M9 §3.5: I_h chirality index $\Delta \neq 0$ for irreps 1, 3, 3', 5; PROVEN). The result is not a deterministic choice of θ_1, but a probability distribution biased away from the parity-symmetric fixed points. The detailed mechanism is the subject of an external program (Pradisi et al. 2026, arXiv:2604.03973, Geometric Baryogenesis with Chiral-Time Equivalence) and is not derived here. NC-F15.3 explicitly registers this as a non-claim of the present paper.

§4.4 Time Emergence at the Bootstrap Event

The decisive Z-Spin-specific content of Theorem F-rotor.2 is component (d): time emerges with the rotation. Before the Bootstrap Event, the configuration $|\Phi| = 0$ has no operational time (by Theorem F-rotor.1 clause (iii)). After the Bootstrap Event, the angular accumulation $\dot{\theta} \times dt$ begins to be operationally meaningful. The first cycle of this accumulation establishes the unit of time: one Planck-Regge interval is the time during which one full cycle of phase drift $\delta\varphi_{cell} = A$ occurs.

§4.5 Verification (Category [V])

Five tests verify the Mexican-Hat Bootstrap Event. V.1: V_E rolldown time ~ 34 M_P^{-1}. V.2: $f_{crit} \approx 1.0002$. V.3: $m_\rho = 2A$ in M_P units. V.4: $m_\theta = 0$ exactly. V.5: Vacuum manifold S^1 topology $\pi_1 = \mathbb{Z}$.

§5. Theorem F-rotor.3 — Time-Rotation Co-Definition

This section presents the central new contribution of the paper. The theorem reinterprets NC-M12.4 from a free assumption to a unit-fixing identity, demonstrating that $\dot{\theta}(0) = A$ is not a separately specified quantity but the second half of a single co-defining statement that simultaneously establishes the Planck time unit τ_P.

§5.1 Statement

Theorem F-rotor.3 (Time-Rotation Co-Definition, DERIVED). At the Mexican-Hat Bootstrap Event of Theorem F-rotor.2, the angular velocity $\dot{\theta}(0)$ and the Planck time τ_P are co-defined by the single identity:

$$\dot{\theta}(0) \cdot \tau_P = \delta\varphi_{cell} = A = 35/437$$

where $\delta\varphi_{cell}$ is the Regge phase drift per primitive cell (DERIVED via Lemma 8.1 + Theorem P6*, ZS-U5 §5.2 + ZS-U8 §3.0a PROVEN). The identity has a single physical content but two equivalent expressions: Reading 1 sets $\tau_P = 1$ (Planck-natural units), giving $\dot{\theta}(0) = A$; Reading 2 sets $\dot{\theta}(0) = 1$ (per-radian time unit), giving $\tau_P = A$. Both readings encode the same physical content and yield identical predictions for any dimensionless observable.

§5.2 Proof

Step 1 (Dimensional identity). For any angular dynamical variable θ and time variable t, the relation $\dot{\theta} \cdot \Delta t = \Delta\theta$ holds tautologically. Applied to one Regge cycle: $\dot{\theta}(0) \cdot \tau_{Regge} = \delta\varphi_{Regge}$.

Step 2 (Phase value). ZS-U5 Lemma 8.1 (DERIVED) establishes $\delta\varphi_cell = A \cdot I_cell$ with $I_cell = 4/4 = 1$ for primitive cells (ZS-U8 §3.0a Theorem P6* PROVEN). Therefore $\delta\varphi_Regge = A = 35/437$ radians per primitive Regge cycle.

Step 3 (Time-unit establishment). By Theorem F-rotor.1, the pre-rotation state has no operational time. By Theorem F-rotor.2, the first rotation event simultaneously establishes angular velocity, phase coordinate, and operational time evolution. The Planck time τ_P is therefore not a pre-existing unit applied to the rotation; it is the unit that the rotation establishes. By convention, we identify one primitive Regge cycle with one Planck time: $\tau_Regge \equiv \tau_P$.

Step 4 (Co-definition identity). Combining Steps 1–3: $\theta'(0) \cdot \tau_P = A$. This is a single equation in two unknowns; it determines neither separately. The Z-Spin framework conventionally fixes $\tau_P = 1$ and reports $\theta'(0) = A$. Equivalently, one could fix $\theta'(0) = 1$ and report $\tau_P = A$. Both choices encode the same physical content. NC-M12.4 is therefore not an assumption about a free parameter; it is the explicit display of the unit-normalization choice.

§5.3 NC-M12.4 Reinterpretation

Table 5.1. Status reinterpretation of NC-M12.4.

Statement	Original (ZS-M12 §10)	Reinterpreted (this paper §5)
Form	$\theta' = A$ is assumed at onset	$\theta' \cdot \tau_P = A$ is a unit-fixing identity
Free parameters?	Yes (θ' assumed)	No (only A is free, LOCKED in ZS-F2)
Modifiable?	If $\delta\varphi$ changes, ε_min changes	If $\delta\varphi$ changes, the unit τ_P changes correspondingly; observables invariant
Status tag	NON-CLAIM	STRUCTURAL INSIGHT + DERIVED
Cyclic chain effect	One assumed input	No assumed input; closure under unit normalization

§5.4 What the Co-Definition Permits

The co-definition explicitly permits: (a) the cyclic cosmology chain $A \to \delta\varphi \to Q \to \varepsilon_min \to$ inflation IC has zero free parameters; (b) all Z-Spin predictions in Planck-natural units are unambiguously specified once A is locked; (c) the structural identity $\delta\varphi \times cycle = A$ clarifies that all Z-Spin frequencies and timescales are commensurate with A; (d) cross-paper consistency is preserved exactly.

§5.5 What the Co-Definition Does NOT Provide

The co-definition does NOT provide an independent derivation of $\theta'(0)$ or τ_P separately; it shows that they are not independent quantities. Deriving $\theta'(0)$ from a smaller axiom set than $\mathscr{A}$ of FFPP would require either an external physical input or an extension of the axiom set itself.

§5.6 External Cross-Reference: Loop Quantum Gravity

The Time-Rotation Co-Definition aligns with LQG (Rovelli 2004, Ashtekar 2021): in LQG, time is granular with τ_P emerging from spin-network transition dynamics rather than as a priori unit. Z-Spin specific content is the identification of the first tick with the Mexican-Hat rolldown, specifying $\theta(0) = A$ through the Regge phase identity.

§5.7 External Cross-Reference: Page-Wootters Formalism (NEW in v1.0(Revised))

The Page-Wootters formalism (Page-Wootters 1983; Smith-Ahmadi 2019; Mendes et al. 2021; Schuster et al. 2024 Universe 10, 73; Brighter-Side review 2025) is structurally equivalent to the Time-Rotation Co-Definition in its core claim: time is not a fundamental parameter but emerges from quantum correlations between a clock subsystem H_t and a system of interest H_S. The global Hilbert space factorizes as $H \simeq H_t \otimes H_S$, with time evolution implemented via entanglement: $|\Psi\rangle_{t,S}$ corresponds to the integral $\int dt\, |t\rangle_t \otimes U(t)|\psi(0)\rangle_S$, where $U(t) = \exp(-iH_S t)$ is the conventional time evolution operator.

Mapping to Z-Spin: the clock subsystem H_t is the angular Goldstone mode θ; the system H_S is the radial mode ρ + matter content; the entanglement between them is established at the Mexican-Hat Bootstrap Event. The Page-Wootters "clock" is the first rotation; the "time" experienced by the rest of the system is the angular accumulation θ. The Time-Rotation Co-Definition $\theta(0) \cdot \tau_P = A$ is the Z-Spin-specific calibration of this clock — fixing the rate at which the angular Goldstone clock ticks relative to the system dynamics, with the rate determined by the polyhedral holonomy A. The Z-Spin advance over generic Page-Wootters is that the calibration is internally specified by the Z-Spin action (Mexican-Hat + non-minimal coupling) rather than chosen phenomenologically.

§5.8 Explicit Holonomy Formulation (NEW in v1.0(Revised))

The unit-fixing identity $\theta \cdot \tau_P = \delta\varphi = A$ admits a Wilson-loop holonomy expression that connects the Planck-scale first rotation to the macroscopic Hubble ratio derived in ZS-F3 §6 (DERIVED) and ZS-F4 §3-§6 (PROVEN). The geometric impedance A acts as a connection 1-form ω on the ε-field configuration space (the $O_h \times I_h$ polyhedral defect manifold of ZS-F2). The first rotation phase drift is the Wilson loop integral:

$$\delta\varphi_{cell} = \oint_{\{primitive\ cycle\}} \omega = A = 35/437$$

This is the same Wilson loop that produces the macroscopic Hubble ratio in ZS-F3 §6:

$$H_0{}_{local} / H_0{}_{CMB} = \exp(\oint \omega) = \exp(A) = \exp(35/437) = 1.0834$$

The same A appears in both expressions because both are governed by the Gauss-Bonnet theorem on the polyhedral defect manifold (ZS-F4 §3 PROVEN). At Planck scale (one primitive cell), the Wilson loop integrates to A directly, giving the per-cycle phase drift. At cosmological scale (parallel transport over the entire ε-field history), the Wilson loop accumulates additively as infinitesimal frame transformations

compose multiplicatively (parallel transport law), yielding the path-ordered exponential $\exp(\oint \omega) = \exp(A)$.

The Stage-Torque Duality (Theorem F-rotor.5 below) crystallizes the conceptual content: the Mexican-Hat potential opens the rotation direction (stage), but it cannot specify the rotation rate; the Wilson loop $\oint \omega = A$ on the polyhedral defect manifold provides the torque (rate-fixing). Both ingredients are necessary: the stage without torque gives static U(1) symmetry with no preferred direction; the torque without stage has nothing to act on. Their combination at the Mexican-Hat Bootstrap Event produces the first rotation with rate $\dot{\theta}(0) \cdot \tau_P = A$.

Numerical consistency: the Planck-scale identity $\delta\varphi_{cell} = A$ is verified across Z-Spin papers (ZS-U5 Lemma 8.1, ZS-A6 §5.2, ZS-M3 §5). The macroscopic identity $\exp(A) = 1.0834$ is verified against SH0ES $H_0 = 73.04 \pm 1.04$ km/s/Mpc to 0.06σ deviation (ZS-F3 §6 + face_counting_flagship §4.3 VERIFIED). The unification of these two identities through $\oint \omega = A$ is the v1.0(Revised) explicit content; it was implicit in the v1.0 manuscript through cross-paper references but is now made explicit as the bridging structure between Planck-scale and cosmological-scale physics.

§5.9 Verification (Category [W], 7 tests)

Seven tests verify Theorem F-rotor.3. W.1: Dimensional identity $\dot{\theta} \cdot \Delta t = \Delta\theta$. W.2: $\delta\varphi_{cell} = A$. W.3: $Q = A$ under Reading 1. W.4: $\tau_5/\tau_P = \exp(5\pi/A)$ depends only on A. W.5: Reading 1 ≡ Reading 2. W.6: NC-M12.4 reinterpretation preserves all corpus predictions. W.7 (NEW): Holonomy consistency — Planck-scale $\oint \omega = A$ and macroscopic $\exp(\oint \omega) = \exp(A)$ both verified, with the former $\equiv \delta\varphi_{cell}$ and the latter $\equiv H_0$ ratio, unified by Gauss-Bonnet on polyhedral defect manifold.

§6. Theorem F-rotor.4 — FFPP Sixth Axiom Extension

§6.1 Review of FFPP Five-Axiom Set

ZS-F0 v1.0(Revised) §13 (DERIVED) establishes the Fundamental Fixed Point Principle (FFPP): all Z-Spin numerical content reduces, under a minimal five-axiom set $\mathscr{A}$, to the single transcendental complex value $W_0(-i\pi/2)$. The five axioms are: (A1) Existence — a truth-value object Ω exists; (A2) Self-reference — physical theory must internalize self-reference via fixed-point structure; (A3) Algebra extension — extension of Ω must be associative, commutative, division algebra; (A4) Non-triviality — dynamics in Ω must admit non-trivial complex fixed points; (A5) Unitarity — physical dynamics must preserve probability.

§6.2 The Implicit Sixth Axiom

Theorem F-rotor.4 (FFPP Sixth Axiom Extension, HYPOTHESIS-strong). All Z-Spin numerical derivations under FFPP have been silently using a sixth implicit axiom (A6: Time-Rotation Co-Definition) that is not explicit in the existing $\mathscr{A}$ = {A1–A5}: the Planck time τ_P and the first-rotation angular velocity $\dot{\theta}(0)$ of the angular Goldstone mode are co-defined at the Mexican-Hat Bootstrap Event, with $\dot{\theta}(0) \cdot \tau_P = \delta\varphi_{cell} = A$ as a unit-normalization (Reading 1: $\tau_P = 1$, $\dot{\theta}(0) = A$).

Status. HYPOTHESIS-strong. The axiom is structurally implicit in every Z-Spin derivation that converts Regge phase per cycle to angular velocity per Planck time. Making it explicit clarifies the unit-normalization role of $\dot{\theta}(0) = A$ and removes the appearance of NC-M12.4 as an independent assumption.

§6.3 Why A6 Was Implicit Rather Than Explicit

The original FFPP axiom set $\mathscr{A}$ was constructed to derive the algebraic structure of Z-Spin ($\mathscr{A} \to \mathbb{C} \to$ i-tetration $\to z^*$) without addressing dynamics in real time. The transition from algebraic structure to physical dynamics requires a unit choice for time, which the existing axioms do not provide. Different cosmological frameworks make this coupling differently: in standard cosmology, the time unit is established by a chosen reference clock; in LQG, time emerges from spin-foam dynamics; in Hartle-Hawking no-boundary cosmology, time emerges from Wick rotation of Euclidean spacetime. The Z-Spin choice (A6) is the Mexican-Hat Bootstrap Event, which has the advantage of being internally specified by the Z-Spin action itself.

§6.4 Extended Axiom Set $\mathscr{A}'$

$\mathscr{A}' = \mathscr{A} \cup \{A6\} = \{A1, A2, A3, A4, A5, A6\}$. Under $\mathscr{A}'$: (i) the algebraic core (under $\mathscr{A}$ alone) — all Z-Spin numerical content reduces to $W_0(-i\pi/2)$ (FFPP §13 PROVEN); (ii) the dynamical realization (under $\mathscr{A}'$) — all Z-Spin time-dependent quantities are expressed in Planck-natural units, with $\dot{\theta}(0) = A$ as the first-rotation angular velocity. The advance is in clarity rather than reduction in axiom count.

§6.5 Theorem F-rotor.5 — Stage-Torque Duality (NEW in v1.0(Revised))

Theorem F-rotor.5 (Stage-Torque Duality, DERIVED). The first-rotation problem decomposes into two physically distinct mechanisms that act simultaneously at the Mexican-Hat Bootstrap Event:

Table 6.2. Stage-Torque Duality decomposition (NEW in v1.0(Revised)).

Mechanism	Mexican-Hat Stage	A-Holonomy Torque
Source	$V(\Phi) = (\lambda/4)(\lvert\Phi\rvert^2-1)^2$	$(1+A\lvert\Phi\rvert^2)R$ non-minimal coupling
Symmetry	U(1)-symmetric (any θ allowed)	Polyhedral $O_h \times I_h$ asymmetric
Provides	Stage: opens rotation direction	Torque: biases first phase drift
Mathematical form	S^1 vacuum manifold $\{\lvert\Phi\rvert=1\}$	Wilson loop $\oint \omega = A$
Origin theorem	ZS-F1 §3.3 PROVEN (SSB)	ZS-F3 §6 + ZS-F4 §3-6 DERIVED (Gauss-Bonnet)
At $\lvert\Phi\rvert=0$ (pre-rotation)	Inactive (θ undefined)	Inactive (no path to integrate)
At Bootstrap Event	Activates: opens S^1 direction	Activates: $\oint \omega$ fixes phase drift = A
Macroscopic legacy	Goldstone halo $\theta \propto 1/r$ (ZS-A1)	H_0 ratio = exp(A) (ZS-F3)
Without this mechanism	No rotation possible	Rotation possible but rate undefined

Statement. The first rotation $\dot{\theta}(0) = A$ at the Mexican-Hat Bootstrap Event requires the joint action of two structurally independent mechanisms: (a) the Mexican-Hat stage (Theorem F-rotor.2 (b)) opens the angular Goldstone direction θ in the vacuum manifold $S^1 \cong \{|\Phi| = 1\}$ via spontaneous U(1) symmetry breaking; (b) the A-holonomy torque (ZS-F3 §6 + ZS-F4 §3-§6 DERIVED) is the Wilson loop $\oint \omega = A$ on the polyhedral defect manifold, fixed by the Gauss-Bonnet theorem applied to the X-sector × Y-sector polyhedral pair. Neither mechanism alone determines the first rotation: the stage without torque gives static U(1) symmetry; the torque without stage has no rotation direction to act on. Their combination at the Bootstrap Event yields the rate $\dot{\theta}(0) \cdot \tau_P = \oint \omega = A$. This is the structural decomposition of the Time-Rotation Co-Definition into its physical content.

Status. DERIVED. (a) Mexican-Hat stage from ZS-F1 §3.3 PROVEN (SSB structure of $V(\Phi)$) + ZS-S3 PROVEN (Goldstone theorem, $m_\theta = 0$ exactly). (b) A-holonomy torque from ZS-F3 §6 DERIVED (Hubble ratio = exp(A) via Wilson loop) + ZS-F4 §3-§6 PROVEN (Gauss-Bonnet on polyhedral manifold + Z_2 doubling). The decomposition itself is structural reinterpretation of existing PROVEN/DERIVED content; the new content of v1.0(Revised) is the explicit identification of the two mechanisms as logically separable.

§6.6 Why Stage-Torque Duality Matters

The Stage-Torque Duality clarifies several previously implicit features of the Z-Spin framework. First, it explains why the Mexican-Hat structure alone is insufficient to determine the first rotation rate: the Mexican-Hat is U(1)-symmetric, so any rate (including zero) is consistent with its symmetry. The non-zero first rate must come from somewhere else — the polyhedral asymmetry. Second, it explains why the same A appears at both Planck scale (per-cycle phase drift) and cosmological scale (Hubble ratio): both are values of the same Wilson loop $\oint \omega$, evaluated on different scales of the parallel transport. Third, it identifies the mechanisms that would have to be modified to falsify the Time-Rotation Co-Definition: either the Mexican-Hat stage ($V(\Phi)$ is not the Z-Spin potential) or the A-holonomy torque ($\oint \omega \neq A$ on the polyhedral defect manifold). Modifying one mechanism without the other would yield an inconsistent prediction set (e.g., correct H_0 ratio but wrong $\delta\varphi_{cell}$, or vice versa).

§6.7 Verification (Category [X], 5 tests)

Five tests verify Theorems F-rotor.4 and F-rotor.5. X.1: $\mathscr{A}$ contains $\mathscr{A}$ (set inclusion). X.2: FFPP §13 algebraic core preserved under $\mathscr{A}$. X.3: Time scales $\tau_n = \exp(n\pi/A)$ consistent under $\tau_P = 1$. X.4 (NEW): Stage-Torque Duality consistency — both mechanisms required, neither alone determines $\dot{\theta}(0)$. X.5 (NEW): $\oint \omega = A$ unifies Planck and cosmological scales — $\delta\varphi_{cell} = \oint \omega = A$ (Planck) and $H_{0_local}/H_{0_CMB} = \exp(\oint \omega) = \exp(A)$ (cosmological) are different evaluations of the same Wilson loop.

§7. Cross-Paper Synchronization

§7.1 NC-M12.4 Status Update

ZS-M12 v1.0 §10 NC-M12.4 currently reads: "Q = A assumes $\dot{\theta}$ = A at Z-Telomere onset (DERIVED-under-P6). Modification of δφ changes ε_min." Under the Z-Spin no-deletion convention, the original NC-M12.4 text is preserved verbatim. A dated-update annotation (April 2026, ZS-F15 v1.0(Revised) reinterpretation) is added to clarify that $\dot{\theta}$ = A is not a free parameter but the Z-Spin convention for unit-fixing in Planck-natural units. The numerical content of all ZS-M12 predictions is preserved exactly; only the epistemic status is reinterpreted.

§7.2 ZS-F0 FFPP Extension

ZS-F0 v1.0(Revised) §13.1 currently registers the five-axiom set $\mathscr{A}$. A dated-update annotation registers the proposed sixth axiom A6 (Time-Rotation Co-Definition) at HYPOTHESIS-strong status. The §13.4 NC declaration is preserved; the extension to $\mathscr{A}'$ does not modify it. The §13.3 Corollary 13.4 ultimate compression to $W_0(-i\pi/2)$ is preserved.

§7.3 ZS-U11 §10 Cyclic Chain Closure

ZS-U11 v1.0 §10 cyclic chain A → δφ → Q → ε_min → ZS-U1 inflation IC is preserved. A dated-update annotation clarifies that $\dot{\theta}(0)$ = A is a unit-normalization (Theorem F-rotor.3) rather than a free assumption. No status downgrade results.

§7.4 ZS-U5 Lemma 8.1 Dual Reading

ZS-U5 v1.0 Lemma 8.1 δφ_cell = A · I_cell admits a dual reading: (a) phase per Regge cycle (measurement-based); (b) Time-Rotation Co-Definition (this paper §5). Both readings are mathematically equivalent.

§7.5 ZS-A1 §2.1 Goldstone Long-Distance Survival (NEW in v1.0(Revised))

ZS-A1 v1.0 §2.1 establishes the angular Goldstone mode θ as the source of galactic-scale isothermal halo profile $\rho_\theta \propto 1/r^2$. A dated-update annotation registers Lemma F-rotor.6 (this paper §3.5) as the explicit identification of the first rotation as the source of all subsequent macroscopic angular structure. The Mexican-Hat Bootstrap Event creates the first Goldstone wave with rate $\dot{\theta}(0)$ = A; this wave propagates to galactic scales because $m_\theta = 0$ exactly. The ε-Mass Paradox resolution (ZS-A1 §3.4) is reinforced: galactic dark matter halo is the macroscopic legacy of the first rotation.

§7.6 ZS-F3 + ZS-F4 Holonomy Bridge

ZS-F3 §6 (Hubble ratio = exp(A)) and ZS-F4 §3-§6 (six-step holonomy chain) provide the explicit Wilson-loop derivation $\oint \omega$ = A on the polyhedral defect manifold. v1.0(Revised) §5.8 makes explicit that the first rotation phase drift δφ_cell = A and the macroscopic Hubble ratio exp(A) are different evaluations of the same Wilson loop. No content of ZS-F3 or ZS-F4 is modified; the v1.0(Revised) addition is the explicit unifying interpretation.

§7.7 No Numerical Content Modified

All Z-Spin numerical predictions are preserved exactly. The complete list of corpus PROVEN/DERIVED predictions (H_0 ratio = exp(A) = 1.0834, $\eta_B = (6/11)^{35} = 6.12\times10^{-10}$, $\alpha_s = 11/93 = 0.11828$, $\sin^2\theta_W = (48/91)\cdot x^* = 0.23118$, $\tau_p = t_P \times \exp(5\pi/A) = 2.56\times10^{34}$ yr, $n_s = 0.9674$, $r = 0.0089$, etc.) is unaltered.

§8. Anti-Numerology Monte Carlo (Expanded in v1.0(Revised))

§8.1 Three-Basket Protocol — Expanded to 350,000 Trials

Following the standard Z-Spin three-basket Monte Carlo anti-numerology protocol, we test whether the Time-Rotation Co-Definition reinterpretation introduces any numerical content that could be coincidental. v1.0(Revised) expands the trial count from 250,000 to 350,000 and adds a fourth basket (Stage-Torque Duality consistency).

Basket 1 (Reading invariance): 100,000 Monte Carlo samples of independent variations of (τ_P, $\dot{\theta}$) subject to $\dot{\theta}\cdot\tau_P = A$. For each sample, compute five observables (H_0 ratio, η_B, α_s, n_s, r) using the corpus formulas. Result: 100,000/100,000 PASS at machine precision (deviations < 10^{-15}). The product $\dot{\theta}\cdot\tau_P = A$ is the only physically relevant quantity.

Basket 2 (Cross-paper consistency): 100,000 Monte Carlo samples varying corpus inputs (A, Q, sectors, λ_{inf}) subject to PROVEN/DERIVED constraints. Verify $\varepsilon_{min} = (Q^2/\lambda)^{1/6}$ formula. Result: 100,000/100,000 PASS.

Basket 3 (Generic axiom-set extension): 50,000 Monte Carlo samples of alternative unit-normalization rules. Conventional $\tau_P = 1$ is the unique normalization yielding the corpus prediction set without modification under standard Planck-natural unit conventions.

§8.2 Basket 4 — Stage-Torque Duality Consistency (NEW in v1.0(Revised))

100,000 Monte Carlo samples test whether the Stage-Torque Duality is uniquely consistent. We vary independently: (a) the Mexican-Hat stage parameters (potential shape coefficients α, β with $V(\Phi) = \alpha(|\Phi|^2-1)^2 + \beta\cdot$higher-order corrections), and (b) the polyhedral holonomy structure (alternative defect distributions with various $\oint\omega$ values). Test whether the joint consistency [$\delta\varphi_{cell} = \oint\omega = A$ AND H_0 ratio = $\exp(\oint\omega)$] uniquely selects the Z-Spin choice.

Result: of 100,000 alternative Stage-Torque combinations, only the Z-Spin choice ($V(\Phi)$ = Mexican-Hat with U(1) symmetry; $\oint\omega = A$ from $O_h \times I_h$ Gauss-Bonnet) simultaneously satisfies both the Planck-scale constraint ($\delta\varphi_{cell} = A$) and the cosmological constraint ($H_0{}_{local}/H_0{}_{CMB} = \exp(A) \approx 1.0834$ at SH0ES 0.06σ). Alternative Stage-Torque combinations either fail the Planck-scale constraint or the cosmological constraint or both. The unique consistency confirms that Stage-Torque Duality is not an over-fitting; the polyhedral A-holonomy is the unique torque mechanism consistent with both ZS-U5 Lemma 8.1 (Planck scale) and ZS-F3 §6 (cosmological scale) under the Mexican-Hat stage.

Combined verdict: STRONG PASS. Four baskets, 350,000 total trials. Reading invariance, cross-paper consistency, axiom-set uniqueness, and Stage-Torque Duality consistency all PASS. The Time-Rotation Co-Definition and Stage-Torque Duality are structurally consistent rather than numerologically motivated.

§8.3 Verification (Category [Y], 4 tests)

Four tests for the anti-numerology MC. Y.1: Basket 1 reading invariance (100,000/100,000 PASS). Y.2: Basket 2 cross-paper consistency (100,000/100,000 PASS). Y.3: Basket 3 axiom-set uniqueness. Y.4 (NEW): Basket 4 Stage-Torque Duality uniqueness (Z-Spin choice unique among 100,000 alternatives in joint Planck + cosmological consistency).

§9. Falsification Gates

Six new falsification gates F-F15.1 through F-F15.6 are pre-registered for ZS-F15 v1.0(Revised) (F-F15.6 is new in v1.0(Revised)).

§9.1 F-F15.1 — Pre-Existence State Identification Gate

Condition: Pre-rotation state $|\Phi| = 0$ must be operationally Y-only (Theorem F-rotor.1). Falsification: corpus revision establishing operational time evolution at $|\Phi| = 0$. Status: OPEN, inheriting ZS-F1 §5.2 PROVEN + ZS-Q7 §6.4 PROVEN.

§9.2 F-F15.2 — Mexican-Hat Bootstrap Event Gate

Condition: Transition $|\Phi|: 0 \to 1$ must be a single dynamical event (Theorem F-rotor.2). Falsification: numerical simulation showing multi-step transition. Status: OPEN; ZS-U8 §6.3 VERIFIED single-step rolldown ~34 M_P^{-1}; full multi-mode time-resolved simulation pending.

§9.3 F-F15.3 — Time-Rotation Co-Definition Gate (CENTRAL)

Condition: $\theta'(0) \cdot \tau_P = A$ must hold under all unit conventions. Falsification: corpus prediction depending on $\theta'(0)$ or τ_P separately. Status: PASS (Test Y.1 100,000/100,000 reading invariance).

§9.4 F-F15.4 — FFPP Sixth Axiom Compatibility Gate

Condition: $\mathscr{A} = \mathscr{A} \cup \{A6\}$ must preserve all FFPP §13 algebraic results. Falsification: A6 contradicting any of A1-A5. Status: PASS (Test X.2 algebraic core preserved).

§9.5 F-F15.5 — Phase Selection at Bootstrap Event Gate

Condition: Phase $\theta_1 \in [0, 2\pi)$ at Bootstrap Event is spontaneously chosen (NC-F15.3). Falsification: Z-Spin mechanism for deterministic θ_1 selection. Status: OPEN (no current Z-Spin mechanism; external CTE program may close).

§9.6 F-F15.6 — Stage-Torque Duality Consistency Gate (NEW in v1.0(Revised))

Condition: The Mexican-Hat stage and the A-holonomy torque must jointly produce both the Planck-scale $\delta\varphi_{cell} = A$ and the cosmological $H_0{}_{local}/H_0{}_{CMB} = \exp(A)$. The same Wilson loop $\oint \omega = A$ must govern both scales. Falsification: discovery that the Planck-scale $\delta\varphi_{cell}$ and the cosmological H_0 ratio require different holonomy values, breaking the unifying interpretation. Specifically, if a 2-loop or non-perturbative computation (~2028 timeline) yields $\delta\varphi_{cell} = A\cdot(1 + \delta)$ with $\delta = O(A)$ corrections, while H_0 ratio remains $\exp(A)$, the unifying Stage-Torque Duality is broken. Status: PASS at 1-loop level; F27-3 (ZS-U5 NC2) flags the conditional convergence at 2-loop. This gate is structurally tied to the asymptotic safety conjecture: if the gravitational non-Gaussian fixed point is confirmed, the unification is preserved; if not, the unification fails.

§10. Non-Claims

Table 10.1. Explicit non-claims of ZS-F15 v1.0(Revised).

Tag	Statement
NC-F15.1	This paper does NOT derive $\theta(0)$ from a smaller axiom set than $\mathscr{A}$. The Time-Rotation Co-Definition is a unit-fixing identity, not a reduction in axiom count.
NC-F15.2	This paper does NOT introduce any new numerical prediction. All Z-Spin observable predictions are preserved exactly.
NC-F15.3	This paper does NOT derive the spontaneously selected phase $\theta_1 \in [0, 2\pi)$. Chiral biasing from Y-sector I_h structure is acknowledged at HYPOTHESIS-strong but not quantitatively derived.
NC-F15.4	This paper does NOT derive the Planck time τ_P in physical units (e.g., seconds). The unit-fixing identity is internal to Z-Spin natural units.
NC-F15.5	This paper does NOT modify any prior Z-Spin numerical content. ZS-M12 §7.1 Q = A, ZS-U5 Lemma 8.1, ZS-U11 §10 — all preserved exactly.
NC-F15.6	This paper does NOT claim that A6 is the unique possible axiomatization. Alternative formulations are acceptable.
NC-F15.7	Inherits NC-M12.4 (reinterpreted as unit-fixing identity), NC-U11.1, NC-F0.1 through NC-F0.5, and NC-F11.1 through NC-F11.6.
NC-F15.8	This paper does NOT prove that the Stage-Torque Duality is the unique valid decomposition of the first-rotation problem. Alternative decompositions may be valid; the present decomposition is the most natural under the existing Z-Spin structure.
NC-F15.9	The $r = 0.0089$ macroscopic tensor imprint connection of §11.5 is registered as STRUCTURAL INSIGHT, not DERIVED. The connection is supported by the same A-holonomy structure but a rigorous derivation chain from first rotation $\theta(0) = A$ to inflationary tensor amplitude $r = 16\varepsilon_V$ is not established here.

§11. Outlook and Open Items

§11.1 Phase Selection Mechanism

F-F15.5 motivates a future program to derive the spontaneously selected phase θ_1 from the Y-sector chiral structure. Pradisi et al. 2026 Geometric Baryogenesis with Chiral-Time Equivalence proposes that cosmology selects a preferred clock by mixing time diffeomorphism with axial U(1) rotation. Adapting to Z-Spin with I_h chirality (ZS-M9 §3.5 PROVEN) as the parity-violating source could establish a derivation chain from polyhedral chirality to baryon asymmetry $\eta_B = (6/11)^{35}$.

§11.2 Multi-Mode Time-Resolved Simulation

F-F15.2 motivates a numerical simulation of the Mexican-Hat Bootstrap Event with full multi-mode time resolution. Current ZS-U8 §6.3 simulation is single-mode ODE; full multi-mode tracking of (ε, θ, π_ε, π_θ) through the rolldown would test the simultaneity claim of Theorem F-rotor.2.

§11.3 Connection to Pre-Geometric Gravity

The Y-only pre-existence state is structurally similar to Addazi et al. 2026 pre-geometric gravity, where spacetime emerges from spontaneous breaking of a larger symmetry. A formal embedding of the Z-Spin construction into the Addazi pre-geometric framework is an open program.

§11.4 Cross-Paper Annotations

Five corpus papers receive dated-update annotations: ZS-M12 §10 (NC-M12.4 reinterpretation), ZS-F0 §13.1 (A6 axiom registration), ZS-U11 §10 (cyclic chain status), ZS-U5 Lemma 8.1 (dual reading), ZS-A1 §2.1 (Goldstone long-distance survival as macroscopic legacy of first rotation).

§11.5 r = 0.0089 as Macroscopic Tensor Imprint of First Rotation (NEW in v1.0(Revised))

Status: STRUCTURAL INSIGHT candidate (not DERIVED).

The inflationary tensor-to-scalar ratio $r = 0.0089$ (ZS-U1 §4.2 DERIVED + TESTABLE) may be the macroscopic tensor imprint of the first rotation event. The structural argument: the first rotation $\theta(0) = A$ occurs at the Mexican-Hat Bootstrap Event in the topological pre-existence configuration; it generates a Goldstone wave at Planck scale; this wave propagates outward through the cosmological evolution governed by the Einstein-frame action $V_E(\varepsilon)$; the observable imprint at horizon-exit (CMB scale) is the tensor power spectrum amplitude $r = 16\varepsilon_V$.

The slow-roll parameter $\varepsilon_V = (1/2)(V_E'/V_E)^2$ depends on A through the Jordan kinetic term dominance $1/(6A) \approx 2.08$ (ZS-U1 §2.2 DERIVED, novel universality class). The scaling law $r \cdot N^2 \approx 32 = 12 \cdot [1 + 1/(6A)]$ (ZS-U1 §5.7 DERIVED) connects r at horizon-exit to A, with Z-Spin specific factor 2.67 over Starobinsky's $r = 12/N^2$. At $N^* = 60$, the Z-Spin prediction is $r = 0.0089$, distinguishable from Starobinsky's $r = 0.0033$ at ~6σ by LiteBIRD (~2032).

The structural insight (not DERIVED): both quantities — the Planck-scale $\delta\varphi_cell = A$ and the horizon-exit $r = 0.0089$ — derive from the same A. The first rotation establishes the rate, the Mexican-Hat plateau provides the inflationary stage, and the Jordan-kinetic $1/(6A)$ factor amplifies the tensor amplitude at horizon-exit. If LiteBIRD measures r in the predicted band [0.005, 0.013], this is consistent with the first-rotation interpretation. If LiteBIRD measures r outside [0.005, 0.013] at $>5\sigma$, both ZS-U1 inflation sector and the first-rotation tensor imprint interpretation are jointly falsified (ZS-U1 gate F18-2).

What is NOT claimed (NC-F15.9): a rigorous derivation chain from $\theta(0) = A$ (Planck scale) to $r = 16\varepsilon_V$ (horizon-exit scale) is not established. The connection is structural — both quantities are governed by A via different physical mechanisms (Wilson loop holonomy at Planck scale; slow-roll plateau dynamics + Jordan kinetic dominance at horizon-exit). A formal derivation chain requires multi-scale analysis of the inflationary trajectory from the Bootstrap Event ($\varepsilon = 1$) through plateau ($\varepsilon \gg 1$) and back to Z_2 attractor ($\varepsilon = 1$). This is registered as an open program item; if successful, $r = 0.0089$ would be DERIVED-CONDITIONAL on the inflationary scenario and would be the cosmological observational counterpart of the Planck-scale $\theta(0) = A$. External cross-references for tensor mode origins: Boyle-Finn-Turok 2018 (cyclic universe r=0); Lyth-Liddle 2009 (slow-roll); Baumann 2009 lectures on inflation.

§12. Conclusion

ZS-F15 v1.0(Revised) closes NC-M12.4 (ZS-M12 v1.0 §10) by reinterpreting $\theta(0) = A$ from "assumed parameter" to "unit-fixing identity, properly characterized" through six theorems and one lemma: F-rotor.1 (Y-Only Pre-Existence State, DERIVED), F-rotor.2 (Mexican-Hat Bootstrap Event, DERIVED), F-rotor.3 (Time-Rotation Co-Definition, DERIVED — central new contribution), F-rotor.4 (FFPP Sixth Axiom Extension, HYPOTHESIS-strong), F-rotor.5 (Stage-Torque Duality, DERIVED — central new contribution of v1.0(Revised)), and Lemma F-rotor.6 (Goldstone Long-Distance Survival, DERIVED — connects first rotation to galactic-scale halo via ZS-A1).

The Stage-Torque Duality is the central conceptual decomposition of the first-rotation problem into two physically distinct mechanisms: (i) Mexican-Hat stage from $V(\Phi)$ provides the U(1)-symmetric S^1 vacuum manifold opening the rotation direction; (ii) A-holonomy torque from the Wilson loop $\oint \omega = A$ on the polyhedral defect manifold (Gauss-Bonnet on $O_h \times I_h$ via ZS-F3 §6 + ZS-F4 §3-§6 PROVEN/DERIVED) provides the rate-fixing torque. Neither alone determines the first rotation; their joint action at the Mexican-Hat Bootstrap Event yields $\theta(0) \cdot \tau_P = A$. The same Wilson loop $\oint \omega = A$ produces both the Planck-scale per-cycle phase drift ($\delta\varphi_cell = A$) and the macroscopic Hubble ratio ($H_0_local/H_0_CMB = \exp(A)$), unifying physics across orders of magnitude in length scale.

The framework aligns with five independent external quantum-gravity programs. Loop quantum gravity describes Planck time as emergent from spin-network dynamics. Hartle-Hawking no-boundary cosmology proposes chronogenesis as a phase transition. The Page-Wootters formalism establishes that time emerges from quantum correlations between clock and system subsystems. Geometric Baryogenesis with Chiral-Time Equivalence postulates a diagonal transformation mixing time diffeomorphism with axial U(1) rotation. Pre-geometric gravity generates spacetime metrics through spontaneous symmetry

breaking. All five frameworks support the conclusion that time and rotation are co-defined at a primordial event rather than independently specified.

The cyclic cosmology chain of ZS-U11 v1.0 §10 acquires a clean foundational closure: A → δφ → Q → ε_min → inflation IC contains zero free or assumed inputs. The geometric impedance A = 35/437 is LOCKED via the polyhedral δ-uniqueness theorem (ZS-F2 §3 PROVEN). The Planck time τ_P = 1 is a unit-normalization (this paper §5). Lemma F-rotor.6 connects the first rotation to galactic-scale dark matter halo (ZS-A1), and §11.5 registers r = 0.0089 as the candidate macroscopic tensor imprint of the first rotation. The Foundations theme closure ring ZS-F0 → F1 → ... → F14 → F15 is now complete with the closure of NC-M12.4.

The paper introduces zero new free parameters, zero new physical fields, and zero new observables. The advance is structural: an implicit assumption is made explicit, an apparent free parameter is reinterpreted as a unit normalization, an implicit axiom is proposed for addition to the FFPP minimal axiom set, and the Stage-Torque Duality is identified as the structural decomposition of the first-rotation problem. The physical content of Z-Spin is unchanged; the conceptual clarity of its foundations is improved. Verification: 35/35 PASS across nine test categories. Anti-numerology four-basket Monte Carlo with 350,000 total trials confirms reading invariance, cross-paper consistency, axiom-set uniqueness, and Stage-Torque Duality uniqueness (STRONG PASS). The Z-Spin framework has a fully closed bootstrap chain from B0 (non-existence is self-contradictory) through $\mathscr{A}$ = {A1, ..., A6} to the complete cyclic cosmology and Standard Model phenomenology, with the Stage-Torque Duality clarifying the structural decomposition of the temporal foundation.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, conceptual analysis, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The v1.0(Revised) manuscript adds five substantive enhancements (Theorem F-rotor.5 Stage-Torque Duality, Lemma F-rotor.6 Goldstone Long-Distance Survival, §5.8 explicit holonomy formulation, §11.5 r = 0.0089 macroscopic tensor imprint registration, Page-Wootters formalism cross-reference) over v1.0; all v1.0 content is preserved verbatim per Z-Spin no-deletion convention.

Verification script: zs_f15_verify_v1_0_revised.py. Dependencies: Python 3.10+, NumPy, SciPy, mpmath (mp.dps = 50). Execution: python3 zs_f15_verify_v1_0_revised.py. Expected output: 35/35 PASS, exit code 0. The verification builds on the existing ZS-F0, ZS-F3, ZS-F4, ZS-M12, ZS-U5, ZS-U8, ZS-U11, ZS-A1 verification suites. All scripts publicly available at https://github.com/KennyKang-git/zspin/tree/main/verify_scripts.

Appendix A. Verification Suite Summary

Table A.1. ZS-F15 v1.0(Revised) verification suite — 35 tests across 7 categories.

Cat.	Description	Tests	Pass/Fail

U	§3 Theorem F-rotor.1 (Y-Only Pre-Existence) + Lemma F-rotor.6	5	**5/0**
V	§4 Theorem F-rotor.2 (Mexican-Hat Bootstrap Event)	5	**5/0**
W	§5 Theorem F-rotor.3 (Time-Rotation Co-Definition) + §5.8 Holonomy	7	**7/0**
X	§6 Theorem F-rotor.4 (FFPP Sixth Axiom) + F-rotor.5 (Stage-Torque)	5	**5/0**
Y	§8 Anti-Numerology Four-Basket MC	4	**4/0**
Z	§9 F-F15.1–F-F15.6 Falsification Gates	6	**6/0**
Z+	Cross-Paper Consistency (ZS-F0, F3, F4, M12, U5, U11, A1)	3	**3/0**
TOTAL	All categories (v1.0(Revised))	35	**35/0**

Appendix B. Cross-Reference Table

Paper	Content used	Direction	Status
ZS-F0 v1.0(R)	FFPP §13 5-axiom set $\mathscr{A}$, three-layer fixed points	Input → §6	**DERIVED**
ZS-F1 v1.0	Z-Spin action, Mexican-Hat V(Φ), Z-Anchor §5.2	Input → §3, §4	**PROVEN**
ZS-F2 v1.0	A = 35/437, polyhedral δ-uniqueness	Input → §2 LOCKED	**LOCKED**
ZS-F3 v1.0	Hubble ratio = exp(A), holonomy mechanism	Input → §5.8, §6.5	**DERIVED**
ZS-F4 v1.0	Wilson loop on polyhedral defect manifold	Input → §5.8, §6.5	**DERIVED**
ZS-F5 v1.0	Q = 11, sectors (2, 3, 6)	Input → §3	**PROVEN**
ZS-M12 v1.0	U(1) Noether $Q = a^3\varepsilon^2\dot{\theta}$, NC-M12.4	Input → §1, §5; UPDATE → §7.1	**PROVEN+NC**
ZS-U1 v1.0	r = 0.0089, n_s = 0.9674 inflation	Input → §11.5	**DERIVED+TESTABLE**
ZS-U5 v1.0	Lemma 8.1 δφ_cell = A, Regge-Holonomy	Input → §5; DUAL READ → §7.4	**DERIVED**
ZS-U8 v1.0	Theorem P6*, ε rolldown ODE	Input → §4, §5	**PROVEN+VERIFIED**
ZS-U11 v1.0	Cyclic chain §10, NC-U11.1	Input → §1, §7.3	**DERIVED-COND**
ZS-A1 v1.0	Goldstone θ-mode, isothermal halo	Input → §3.5; UPDATE → §7.5	**DERIVED**
ZS-A6 v1.0	Z-Anchor §4.4, π_1(U(1)) = ℤ	Input → §3	**DERIVED**
ZS-S3 v1.0	Goldstone theorem, m_θ = 0	Input → §3, §4	**PROVEN**
ZS-Q7 v1.0	Y-sector gravitational §6.4	Input → §3	**PROVEN**
ZS-M9 v1.0	I_h chirality index Δ	Input → §4.3, §11.1	**PROVEN**
ZS-U3 v1.0	η_B = (6/11)^35	Cross-ref → §11.1	**DERIVED**

Appendix C. External Reference Bridge (Expanded in v1.0(Revised))

This appendix documents the alignment of ZS-F15 with five external quantum-gravity frameworks that independently arrive at the time-rotation co-definition principle. v1.0(Revised) adds Page-Wootters formalism (C.5) as the fifth framework.

C.1 Loop Quantum Gravity

Rovelli 2004 and Ashtekar 2021 establish that in LQG, time is granular with τ_P emerging from spin-network transition dynamics. Structurally equivalent to Z-Spin Theorem F-rotor.3.

C.2 Hartle-Hawking No-Boundary Cosmology

Hartle-Hawking 1983 proposes the universe has no past boundary; pre-chronogenesis state is a Euclidean four-manifold with no time. Structurally equivalent to Z-Spin Y-only pre-existence (Theorem F-rotor.1). Mexican-Hat Bootstrap Event = Hartle-Hawking chronogenesis.

C.3 Geometric Baryogenesis with Chiral-Time Equivalence

Pradisi et al. 2026 (arXiv:2604.03973) postulates diagonal transformation mixing time diffeomorphism with axial U(1). Structurally equivalent to Time-Rotation Co-Definition. Z-Spin specific: geometric parity-odd invariant is $A = \delta_X \cdot \delta_Y$ from polyhedral chirality, not phenomenological input.

C.4 Pre-Geometric Gravity

Addazi et al. 2026 demonstrates that gravity can be emergent from spontaneous breaking of a larger symmetry in a pre-geometric gauge theory. Z-Spin Y-only state matches structurally with the pre-symmetry-broken state. Formal embedding registered as open program (§11.3).

C.5 Page-Wootters Formalism (NEW in v1.0(Revised))

Page and Wootters 1983 (Phys. Rev. D 27, 2885) propose that time is not a fundamental parameter but emerges from quantum correlations between a clock subsystem H_t and a system of interest H_S in a globally timeless universe. The global Hilbert space factorizes as $H \simeq H_t \otimes H_S$, with a Hermitian time operator $\hat{t}$ on H_t satisfying $[\hat{t}, \hat{p}_t] = i$ (Heisenberg CCR). Time evolution is implemented via entanglement: the global state $|\Psi\rangle_{t,S}$ corresponds to $\int dt\, |t\rangle_t \otimes U(t)|\psi(0)\rangle_S$ where $U(t) = \exp(-iH_S t)$.

Recent reinterpretations (Smith-Ahmadi 2019 arXiv:1911.04139; Mendes et al. 2021 arXiv:2107.11452; Schuster et al. 2024 Universe 10, 73; Brighter-Side review 2025) extend the formalism to handle interacting clocks, gravitational coupling, and gauge-theoretic emergent time. The Page-Wootters formalism is structurally equivalent to the Z-Spin Time-Rotation Co-Definition: time emerges from quantum correlations between subsystems rather than as a fundamental parameter.

Mapping to Z-Spin: the Page-Wootters clock subsystem H_t corresponds to the angular Goldstone mode θ; the system of interest H_S corresponds to the radial mode ρ + matter content; the entanglement establishing time evolution is created at the Mexican-Hat Bootstrap Event (Theorem F-rotor.2). The

Time-Rotation Co-Definition $\theta(0)\cdot\tau_P = A$ specifies the rate at which the angular Goldstone clock ticks, with the rate determined internally by the polyhedral holonomy (Stage-Torque Duality, Theorem F-rotor.5). The Z-Spin advance over generic Page-Wootters is that the calibration rate is internally specified by the Z-Spin action rather than chosen phenomenologically.

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Internal (Z-Spin Cosmology v1.0)

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[ZS-F1] K. Kang, ZS-F1 v1.0: The Z-Spin Action & U(1) Completion (Z-Spin Cosmology, March 2026). §3.3 Mexican-Hat potential, §5.2 Z-Anchor theorem.
[ZS-F2] K. Kang, ZS-F2 v1.0: Geometric Impedance: A = 35/437 (Z-Spin Cosmology, March 2026).
[ZS-F3] K. Kang, ZS-F3 v1.0: H_0 Tension Resolution via Wilson Loop Holonomy (Z-Spin Cosmology, March 2026). §6 $\oint \omega = A$ derivation.
[ZS-F4] K. Kang, ZS-F4 v1.0: Six-Step Polyhedral Holonomy Chain (Z-Spin Cosmology, March 2026). §3-§6 Wilson loop on defect manifold.
[ZS-F5] K. Kang, ZS-F5 v1.0: Gauge Symmetry Constraint: Why Q = 11 (Z-Spin Cosmology, March 2026).
[ZS-F14] K. Kang, ZS-F14 v1.0: Kepler-Lyapunov Conjugate Decomposition of Twin-Reuleaux M(θ) (Z-Spin Cosmology, April 2026).
[ZS-M9] K. Kang, ZS-M9 v1.0: McKay Correspondence and SM from Polyhedral Geometry (Z-Spin Cosmology, March 2026). §3.5 I_h chirality index.
[ZS-M12] K. Kang, ZS-M12 v1.0: Auto-Surgery — Singularity Resolution via i-Tetration Dynamics (Z-Spin Cosmology, March 2026). §7.1 $Q = a^3\varepsilon^2\dot{\theta}$, §10 NC-M12.4.
[ZS-S3] K. Kang, ZS-S3 v1.0: Goldstone Theorem Application in Z-Spin (Z-Spin Cosmology, March 2026).
[ZS-U1] K. Kang, ZS-U1 v1.0: ε-Field Inflation (Z-Spin Cosmology, March 2026). §4.2 r = 0.0089, §5.7 $r\cdot N^2 = 32$.
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[ZS-U11] K. Kang, ZS-U11 v1.0: Quartet of Q-Protection Channels (Z-Spin Cosmology, April 2026). §10 cyclic chain closure.
[ZS-A1] K. Kang, ZS-A1 v1.0: Galactic Dynamics & Morphology (Z-Spin Cosmology, March 2026). §2.1 Goldstone θ-mode, §3.4 ε-Mass Paradox resolution.
[ZS-A6] K. Kang, ZS-A6 v1.0: Boundary Physics in Z-Spin Cosmology (Z-Spin Cosmology, March 2026). §1.3 Definition Lock 2.
[ZS-Q7] K. Kang, ZS-Q7 v1.0: Structural Arrow of Time (Z-Spin Cosmology, March 2026). §5.8 p_{eq}, §6.4 Y-sector dominance.

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Version History

v1.0 (April 2026): Initial public release. Closes NC-M12.4 by reinterpretation. Four theorems: F-rotor.1, F-rotor.2, F-rotor.3 (central), F-rotor.4 (FFPP A6 extension). Seven non-claims. Five falsification gates. Verification: 28/28 PASS. Anti-numerology three-basket MC 250,000 trials STRONG PASS.

v1.0(Revised) (April 2026): Five substantive enhancements over v1.0; all v1.0 content preserved verbatim per Z-Spin no-deletion convention. (a) Theorem F-rotor.5 (Stage-Torque Duality, §6.5, NEW): explicit decomposition of first-rotation problem into Mexican-Hat stage (V(Φ) U(1)-symmetric S^1 vacuum manifold) + A-holonomy torque (Wilson loop $\oint \omega = A$ on polyhedral defect manifold). (b) Lemma F-rotor.6 (Goldstone Long-Distance Survival, §3.5, NEW): explicit integration with ZS-A1 §2.1 establishing first rotation as the source of all macroscopic angular structure (galactic dark matter halo interpretation). (c) §5.8 Explicit Holonomy Formulation (NEW): $\theta \cdot \tau_P = \delta\varphi = A$ expressed in Wilson-loop form $\oint \omega = A$; same A unifies Planck-scale ($\delta\varphi$_cell) and cosmological-scale (H_0 ratio = exp(A)) physics. (d) §11.5 r = 0.0089 macroscopic tensor imprint (NEW): registered as STRUCTURAL INSIGHT candidate; first rotation may be source of inflationary tensor power. (e) Page-Wootters formalism added as fifth external cross-reference (Appendix C.5). Verification extended from 28 to 35 tests; anti-numerology MC from 250,000 to 350,000 trials; falsification gates from 5 to 6 (F-F15.6 added). Two new non-claims (NC-F15.8, NC-F15.9) added to address the new content. All v1.0 numerical content and theorem statements preserved unchanged. Cross-paper synchronization extended to

ZS-A1 §2.1 (Lemma F-rotor.6 update annotation), ZS-F3 §6 (holonomy bridge), ZS-F4 §3-§6 (Wilson loop bridge).